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QR Decomposition

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LICENSE	Initial commit	5 years ago
README.md	succes	now
Screenshot 2026-05-24 165...	succes	now
eq1.jpg	added experiment	5 years ago
eq2.jpg	added experiment	5 years ago
eq3.jpg	added experiment	5 years ago
ex1.jpg	updated	5 years ago
ex2.jpg	updated	5 years ago
ex3.jpg	updated	5 years ago
ex4.jpg	updated	5 years ago
ex6.jpg	updated	5 years ago
input.jpg	added experiment	5 years ago

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Algorithm for QR Decomposition

Aim:

To implement QR decomposition algorithm using the Gram-Schmidt method.

Equipment's required:

1. Hardware – PCs
2. Anaconda – Python 3.7 Installation / Moodle-Code Runner

Algorithm:

1. Intialize the matrix Q and u
2. The vector u and e is given by

$$u_1 = a_1$$

$$u_2 = a_2 - (a_2.e_1)e_1$$

$$u_3 = a_3 - (a_3.e_1)e_1 - (a_3.e_2)e_2$$

$$e_1 = \frac{u_1}{\|u_1\|}$$

$$e_2 = \frac{u_2}{\|u_2\|}$$

3. Obtain the Q matrix

$$Q = (e_1|e_2| \dots \dots \dots e_n)$$

4. Construct the upper triangular matrix R

$$R = \begin{bmatrix} a_1.e_1 & a_2.e_1 & a_3.e_1 \\ 0 & a_2.e_2 & a_3.e_2 \\ 0 & 0 & a_3.e_3 \end{bmatrix}$$

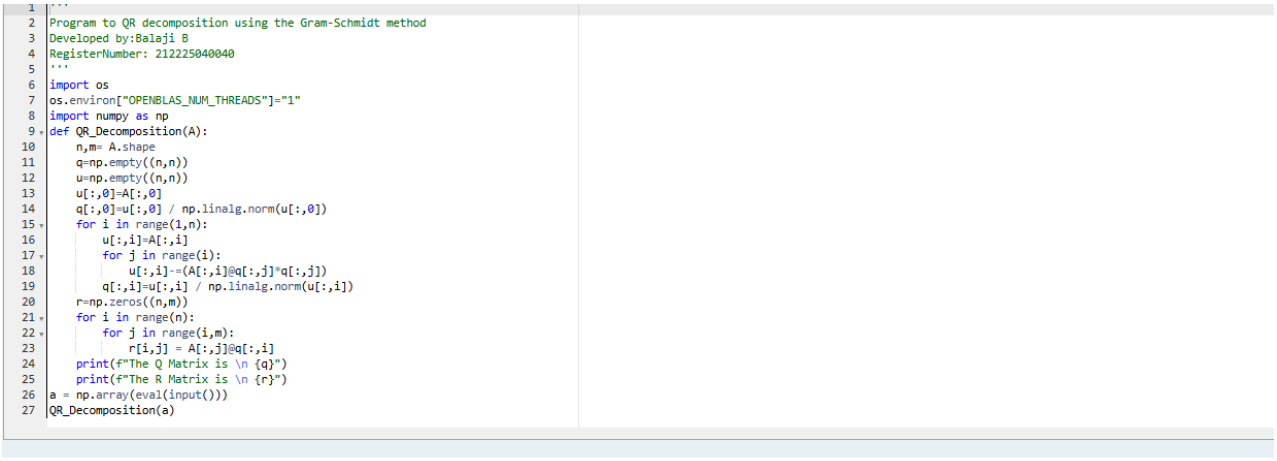
Program:

Gram-Schmidt Method

```
'''
Program to QR decomposition using the Gram-Schmidt method
Developed by:Balaji B
RegisterNumber: 212225040040
'''
import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
import numpy as np
def QR_Decomposition(A):
    n,m= A.shape
    q=np.empty((n,n))
    u=np.empty((n,n))
    u[:,0]=A[:,0]
    q[:,0]=u[:,0] / np.linalg.norm(u[:,0])
```

```
    for i in range(1,n):
        u[:,i]=A[:,i]
        for j in range(i):
            u[:,i]-=(A[:,i]@q[:,j]*q[:,j])
        q[:,i]=u[:,i] / np.linalg.norm(u[:,i])
    r=np.zeros((n,m))
    for i in range(n):
        for j in range(i,m):
            r[i,j] = A[:,j]@q[:,i]
    print(f"The Q Matrix is \n {q}")
    print(f"The R Matrix is \n {r}")
a = np.array(eval(input()))
QR_Decomposition(a)
```

Output



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    for i in range(1,n):
        u[:,i]=A[:,i]
        for j in range(i):
            u[:,i]-=(A[:,i]@q[:,j]*q[:,j])
        q[:,i]=u[:,i] / np.linalg.norm(u[:,i])
    r=np.zeros((n,m))
    for i in range(n):
        for j in range(i,m):
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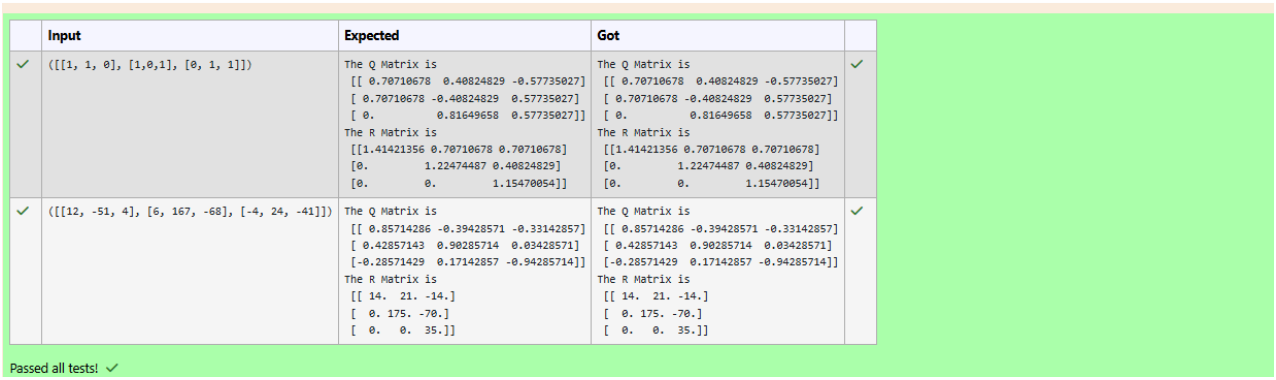
    print(f"The Q Matrix is \n {q}")
    print(f"The R Matrix is \n {r}")
a = np.array(eval(input()))
QR_Decomposition(a)
```



	Input	Expected	Got	
✓	([[1, 1, 0], [1,0,1], [0, 1, 1]])	The Q Matrix is [[0.70710678 0.40824829 -0.57735027] [0.70710678 -0.40824829 0.57735027] [0. 0.81649658 0.57735027]] The R Matrix is [[1.41421356 0.70710678 0.70710678] [0. 1.22474487 0.40824829] [0. 0. 1.15470054]]	The Q Matrix is [[0.70710678 0.40824829 -0.57735027] [0.70710678 -0.40824829 0.57735027] [0. 0.81649658 0.57735027]] The R Matrix is [[1.41421356 0.70710678 0.70710678] [0. 1.22474487 0.40824829] [0. 0. 1.15470054]]	✓
✓	([[12, -51, 4], [6, 167, -68], [-4, 24, -41]])	The Q Matrix is [[0.85714286 -0.39428571 -0.33142857] [0.42857143 0.90285714 0.03428571] [-0.28571429 0.17142857 -0.94285714]] The R Matrix is [[14. 21. -14.] [0. 175. -70.] [0. 0. 35.]]	The Q Matrix is [[0.85714286 -0.39428571 -0.33142857] [0.42857143 0.90285714 0.03428571] [-0.28571429 0.17142857 -0.94285714]] The R Matrix is [[14. 21. -14.] [0. 175. -70.] [0. 0. 35.]]	✓

Passed all tests! ✓

Result

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